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TOLID: Turkish Offensive Language Identification Dataset and Transformer-Based Benchmarks

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WHAT IS ALREADY KNOWN ON THIS TOPIC?

- *Existing studies on Turkish offensive language detection typically rely on small, imbalanced, and topic-restricted datasets and often focus on specific target groups (e.g., women, refugees, ethnic minorities).*
- *Most previous research adopts binary or coarse-grained annotation schemes, limiting the detailed analysis of more nuanced offensive types such as sexist, racist, political, or religious insults.*
- *Although transformer-based models have been applied in Turkish natural language processing, their performance remains constrained due to limited dataset diversity, insufficient annotation quality, and class imbalance.*

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ABSTRACT

This study presents the Turkish Offensive Language Identification Dataset (TOLID), a large-scale, high-quality dataset for the automatic detection of offensive language in Turkish social media posts and evaluates the performance of several transformer-based models. Although most previous studies have been constrained by small sample sizes, imbalanced class distributions, or narrow topical focus, TOLID includes a wide range of offensive expressions without imposing restrictions on topics, individuals, or groups. The dataset was annotated by three independent experts using a hierarchical and fine-grained scheme that addresses not only general offensive language but also specific subcategories such as sexist, racist, political, and religious insults. To evaluate the dataset, several transformer-based models adapted for Turkish, including BERTurk (Bidirectional Encoder Representations from Transformers for Turkish), ConvBERTurk (Convolutional Bidirectional Encoder Representations from Transformers for Turkish), and ELECTRA-Turkish (Efficiently Learning an Encoder that Classifies Token Replacements Accurately for Turkish), were trained and tested. Among these, ConvBERTurk achieved the highest scores, reaching 82.76% macro F1 in offensive vs. non-offensive classification and 78.14% in targeted vs. non-targeted offensive classification, outperforming previous research. These results demonstrate that combining a balanced, multi-annotated dataset with advanced deep learning architectures can effectively address the linguistic richness, contextual complexity, and informal nature of Turkish social media text. Additionally, a web-based application was developed to provide a practical interface for analyzing text and visualizing model outputs, extending the study's impact beyond academic research to real-world applications. Overall, this study addresses key limitations of prior research and makes a significant contribution to Turkish natural language processing by providing a meticulously constructed dataset and extensive benchmarks with state-of-the-art models. TOLID establishes a robust foundation for future work on offensive language subtypes, automatic moderation, and toxicity analysis in Turkish social media.
Index Terms— Deep learning, hate speech, offensive language, text classification, Turkish

I. INTRODUCTION

The rise in user-generated content on online platforms has provided opportunities to expand freedom of expression, while simultaneously creating a breeding ground for the spread of misinformation, insults, hate speech, and violent content. Social media platforms, by allowing users to remain anonymous, often facilitate the dissemination of negative content on social media and may even encourage real-world violence [1, 2]. Consequently, these platforms can become conducive environments for hosting undesirable content [3, 4].

The detection of various forms of offensive language has garnered significant attention from both academic and commercial entities. In addition to academic studies focusing on the linguistic, psychological, and sociological aspects of inappropriate language, practical applications aimed at automatically detecting such language have also emerged. These technologies can be applied in various fields, including content moderation on social media platforms, detection of illegal content by law enforcement, and protection of children from harmful content.

WHAT THIS STUDY ADDS TO THIS TOPIC?

- *This study introduces TOLID, the most comprehensive, balanced, and fine-grained Turkish offensive language dataset to date, annotated by three experts using a hierarchical scheme covering both general and targeted offensive subtypes.*
- *It presents the most extensive comparative evaluation for Turkish offensive language detection and demonstrates that ConvBERTurk surpasses prior work, achieving 82.76% macro-F1 for offensive detection and 78.14% macro-F1 for target identification.*
- *It extends the scientific contribution into practice by developing a web-based application that enables real-time offensive language detection and visualization, turning the models into a practical tool for content moderation and toxicity monitoring.*

In Türkiye, the widespread use of social media—particularly the prevalence of X (formerly Twitter)—provides a valuable source for collecting and analyzing such content [5].

The Turkish language, with its agglutinative structure, flexible word order, and frequent omission of arguments, presents typological differences compared to previous studies that primarily focus on English and other Indo-European languages. Given Türkiye's unique position as a cultural and geographical bridge between East and West, the language and expressions used in Türkiye possess distinctive characteristics. The increasing online censorship and government surveillance in the country have heightened pressure on social media users, leading them to adopt more indirect forms of expression [6, 7]. Additionally, factors such as the presence of Syrian refugees and the government's involvement in the Syrian conflict contribute to societal polarization, influencing the language used in this context [8]. Therefore, within these dynamics, social media platforms must take more effective measures in detecting and managing such negative content.

Social media platforms have established specific rules to minimize undesirable content, and users are expected to adhere to these rules [9].

Posts that violate these rules may be removed, and user accounts may be suspended. Platforms employ moderators who manually review posts to ensure a healthy, reliable, and user-friendly environment [10].

However, this moderation strategy is dependent on factors such as the speed of the moderators, their ability to understand slang and jargon, and their proficiency in multilingual content. Additionally, due to the vast volume of data flow, it is nearly impossible to manually review every post and filter harmful content. Thus, there is a growing need for automatic techniques to detect offensive language. Automatic techniques play a critical and indispensable role in identifying such content. Recent increases in research on offensive language are primarily driven by the following factors:

- **Social Awareness:** Growing societal awareness has heightened interest in combating offensive language. The prevalence of such content on social media platforms has led to significant public concern and reactions.
- **Legal and Political Pressures:** Governments and institutions are implementing regulations and policies to control the spread of offensive language. These pressures encourage researchers to focus on this area.
- **Advancements in Communication Technologies:** The widespread use of the internet and social media has increased communication among people, leading to a rise in content containing offensive language. This has motivated researchers to work on detecting and preventing such content.
- **Technological Progress:** Advances in natural language processing (NLP), machine learning, and artificial intelligence have facilitated the development of automated detection methods for offensive language. These technological capabilities encourage further research in this field.
- **Social Impact**: The negative effects of offensive language have serious consequences for individuals, communities, and organizations. Awareness of these impacts drives people to seek research and solutions to address the issue.
- **Interest from Companies and Social Media Platforms:** Social media platforms and companies prioritize combating offensive language to ensure a safe and positive environment for communication. Consequently, they support and fund research in this area.

These factors have motivated the current study, which aims to develop a reliable system for detecting offensive language in Turkish. To this end, a new dataset was created and annotated by multiple experts, and transformer-based models were trained to achieve state-of-the-art performance in offensive language detection tasks. The contributions are as follows:

- **A More Comprehensive and Balanced Dataset:** While previous studies on offensive and hate speech in Turkish have made notable contributions to the literature, they are often limited by small dataset sizes, imbalanced class distributions, and narrow topical focus. For instance, Şahi et al. [11] focused exclusively on sexist speech targeting women; [12] examined hate speech directed at specific groups (e.g., ethnic and gender groups); and Mayda et al. [13, 14] concentrated on predefined target groups. Similarly, Beyhan et al. [15] limited their analyses to specific topics such as the Istanbul Convention and refugee-related discourse. To address these limitations, the Turkish Offensive Language Identification Dataset (TOLID) was designed as a comprehensive resource that captures diverse forms of offensive and hate speech without restrictions on topics, individuals, or groups. Featuring a detailed hierarchical annotation scheme, TOLID provides a broader and more representative foundation for developing robust and generalizable models in Turkish offensive language detection.

- **Detailed and Fine-Grained Annotation Strategy**: In the literature, offensive language detection is often addressed using limited and superficial classification schemes. Most studies categorize content in a binary manner as "offensive" or "non-offensive," or use three classes such as "offensive speech," "hate speech," and "neither." In contrast, the TOLID dataset adopts a more detailed and comprehensive annotation strategy that goes beyond these traditional approaches. Constructed through the careful and systematic work of three independent annotators, TOLID achieves high consistency despite common challenges highlighted in the literature, such as the lack of a clear definition of hate and offensive speech, subjective interpretations, and context-dependent meaning shifts. Moreover, TOLID does not simply classify offensive speech as a single broad category but further divides it into specific subcategories, including sexist, racist, political, and religious group-directed insults. This fine-grained annotation approach enables detailed analysis of various types of both targeted and untargeted hate speech, providing a valuable resource for researchers working on specific subdomains.
- **Implementation and Comparison of Transformer-Based Approaches:** In recent years, transformer architectures have achieved remarkable success in text classification and have become a standard in many languages. This study provides a comprehensive evaluation of various transformer-based models adapted for Turkish— namely BERTurk (Bidirectional Encoder Representations from Transformers for Turkish), ConvBERTurk (Convolutional Bidirectional Encoder Representations from Transformers for Turkish), and ELECTRA-Turkish (Efficiently Learning an Encoder that Classifies Token Replacements Accurately for Turkish) —investigating their effectiveness in detecting hate speech and offensive language in Turkish social media content. The conducted experiments and comparative analyses demonstrate that modern deep learning approaches can achieve high performance despite the morphological richness and contextual complexity of the Turkish language. Notably, the ConvBERT-based model outperformed previously reported results, achieving a macro F1 score of 82.76% for distinguishing offensive from non-offensive content, and 78.14% for classifying targeted vs. untargeted offensive language. These findings highlight the strong potential of transformer-based models as robust solutions for detecting hate speech and offensive language in Turkish texts.
- **Web-Based Application Development:** A web-based application was designed to visualize the outputs of the developed models and to provide users with an interface to analyze their own texts. Thus, the model has moved beyond being solely an academic study and has become a practical moderation tool for real-world use. Users can perform real-time queries, enabling offensive and hate speech detection to be applied in everyday contexts. In this way, the study offers a concrete contribution to Turkish NLP both scientifically and socially.
- **Contribution to Turkish NLP:** Research on the detection of offensive and hate speech in Turkish remains limited. This study bridges that gap by introducing the meticulously constructed fine-grained TOLID dataset, conducting extensive experiments, and developing high-performance models that together provide a solid foundation for future work in this field. Furthermore, its methodological approach and the resources it offers make a strong and multifaceted contribution to Turkish NLP research.

The remainder of this paper is structured as follows: Section 2 reviews previous studies on offensive language detection in social media, focusing on the scope and limitations of Turkish research. Section 3 presents the creation process and analysis of the TOLID dataset. Section 4 details experiments and performance comparisons of three transformer-based models on Turkish social media data. Section 5 discusses the findings, addresses limitations, and offers suggestions for future work. Finally, Section 6 summarizes the conclusions and highlights the contributions of the dataset and models to Turkish offensive language detection.

II. RELATED WORK

In recent years, the creation and dissemination of negative discourse based on gender, race, ethnicity, religion, and political ideologies through social media have become increasingly common. Consequently, a significant amount of research has focused on detecting unwanted user-generated content, such as profanity, sexism, racism, mocking, trolling, cyberbullying, and offensive or derogatory language.

The rise in studies on the detection of offensive language can be attributed to increased social awareness, legal and political pressures, advancements in communication technologies, and the interest of companies and social media platforms. Offensive language includes threats, discrimination, profanity, and gross insults directed at individuals [16]. Hate speech, on the other hand, refers to language that attacks or demeans a group based on race, ethnicity, religion, disability, gender, age, or sexual orientation [17]. In this study, the term "offensive language" is preferred as a general term that also encompasses hate speech.

This literature review examines existing research on the automatic detection of offensive language from a broad perspective. Researchers have employed a range of machine learning and NLP techniques to detect offensive content across various languages and online platforms. While the majority of studies have been conducted in English, research on offensive language and hate speech detection has also attracted considerable attention in other languages. In one of the first studies, Waseem and Hovy [18] analyzed hate speech in social media, categorizing it into racism and sexism and reporting promising results with character n-grams and gender features (F1 = 0.739). Similarly, Nobata et al. [17] developed a supervised model based on linguistic and syntactic features using a large Yahoo comments dataset of 3.3 million entries, showing that feature engineering could effectively address offensive language detection even without deep learning. Google's Jigsaw division organized Kaggle competitions, providing extensive datasets for detecting toxic comments in English, encouraging participants to develop AI models to classify these texts [19, 20].Davidson et al. [21] classified 24,802 tweets into hate speech, offensive language, and neither, achieving an F1 score of 0.90 with logistic regression and support vector machine (SVM). Poletto et al. [22], focusing on Italian data, investigated hate speech targeting refugees and Muslims and reported a moderate inter-annotator agreement (Cohen's kappa = 0.54). Ross et al. [23] examined German tweets, adopting binary labels for hate speech and graded categories for offensive language, where annotator subjectivity was identified as a major issue. More recently, Zampieri et al. [24] proposed a hierarchical three-level labeling scheme and tested deep learning models such as bidirectional long short-term memory and convolutional neural network (CNN) for English Twitter data, contributing to more fine-grained offensive language classification.

In the context of Turkish, several notable studies have emerged. Şahi et al. [11] analyzed sexist hate speech within 1288 tweets collected under the hashtag #kıyafetimekarışma, identifying 159 sexist instances after annotation filtering. Dağaşan [12] focused on hate speech directed at predefined groups, such as Armenians, Kurds, Syrians, women, and LGBTI+ individuals, within a dataset of 4081 tweets, 884 of which were labeled as hate speech. Çöltekin [25] created the first large-scale Turkish dataset consisting of 34,805 manually labeled tweets, which supported three classifiers targeting offensive speech, targeted speech, and type classification, achieving F1 scores of 0.773, 0.779, and 0.530, respectively; this dataset was later included in SemEval-2020 [26]. Mayda et al. [13] contributed a smaller dataset of 1000 tweets, with 336 labeled as hate or offensive, while their subsequent study [14] expanded this effort to 10,224 tweets targeting 19 groups, resulting in 2,236 hate speech and 166 offensive instances. More recently, Beyhan et al. [15] applied BERTurk to detect hate speech, reporting a binary classification accuracy of 77% with 5-fold cross-validation on the Istanbul Convention dataset and 71% on the Refugee dataset. Additionally, Tanyel et al. [27] addressed bias in hate speech detection by introducing data augmentation techniques and tested their approach on both English and Turkish datasets, demonstrating that such methods can improve model robustness across languages.

Studies on offensive language in Turkish have made notable contributions to the literature by using various datasets and methodologies. Nevertheless, many of these works exhibit significant limitations, including small dataset sizes, imbalanced class distributions, and a focus on specific subjects or groups. For instance, Şahi et al. [11] focused on sexist tweets, working with just 159 instances; Dağaşan [12] concentrated on specific ethnic and gender groups; Mayda et al. [13, 14] examined hate speech targeting predefined groups; and Beyhan et al. [15] examined tweets about refugees and the Istanbul Convention.

In this study, a comprehensive dataset was created with a detailed labeling hierarchy designed to cover different types of offensive language without any subject, person, or group restrictions. While this approach allows for a detailed analysis of various types of offensive language, it is also flexible for research focusing on a specific type of discourse. The annotation process was inspired by the hierarchical method proposed by Zampieri et al. [24] and was largely aligned with the annotation guidelines introduced by Çöltekin [25]. Moreover, to address the low prevalence issue (19.4%) observed in Çöltekin's dataset, a classification algorithm was utilized during the data collection phase to prioritize posts with a high likelihood of containing offensive language. As a result, a more balanced and reliable Turkish dataset containing examples of offensive language across multiple categories has been introduced to the literature.

III. DATASET

To develop a dataset for detecting offensive language in Turkish, existing dataset creation approaches in the literature were examined in detail. Social media content often contains relatively low levels of offensive discourse [28]. Therefore, one of the potential challenges in constructing an offensive language dataset is identifying offensive content among a large volume of non-offensive posts. This issue makes the annotation process both less efficient and more time-consuming [25].

As a response to this challenge, commonly used approaches in the literature were examined. Previous studies have utilized approaches such as keyword-based filtering, monitoring offensive discourse targeting specific groups, and tracking posts from users known to employ offensive language to increase the proportion of offensive content in datasets [18, 24, 29–31]. In the present study, the keyword-based sampling method proposed by Waseem and Hovy [18] was adopted. This allowed for the prioritization of posts with a high likelihood of containing offensive content, thereby improving the efficiency of the annotation process.

A. Data Collection

For the dataset developed in this study, 1.67 million unlabeled X posts collected by Kurt and Yücel [32] using keywords with the potential for offensive content were utilized. Detailed information on the data collection process is provided in the referenced study. From these posts, a random subset of 5202 was manually annotated by three independent annotators in accordance with the annotation guidelines presented in the next section. The annotated dataset will be shared publicly on GitHub, with a flexible license, to contribute to studies on Turkish offensive language detection [33].

B. Data Annotations

This section provides information on the annotation of 5202 X posts preselected in the previous step by three annotators. The annotation process was based on the hierarchical scheme proposed by Zampieri et al. [24] while the label names and guidelines were largely structured in alignment with Çöltekin [25].

The lack of a universally agreed-upon definition of offensive language, the subjectivity of annotations, and the context-dependent nature of meaning are issues frequently emphasized in the literature [17, 21, 24, 25]. This study aimed to minimize such subjective differences and develop a more consistent and high-quality dataset. To this end, a multi-annotator approach was adopted; each post was independently evaluated by three annotators, and the final label was determined based on consensus. If a post was marked as "X" by at least one annotator, it was assigned a final label of "X," indicating that the post was either not in Turkish or was considered "difficult to classify" by at least one annotator.

The annotation process was conducted based on the guidelines shown in Fig. 1, which are based on the recommendations of Çöltekin [25] with minor modifications made to meet the needs of this study. These guidelines aim to systematically categorize different aspects of offensive language and provide researchers with the opportunity to conduct more detailed analyses. For instance, studies focusing on gender-based offensive language can utilize posts labeled "GRP1," whereas those exploring ethnic-based discourse may concentrate on posts marked "GRP2."

Step 1: Is the post in Turkish and understandable?

- **Yes:** Continue to the next step
- **No:** Select **X** label and move to next post

Step 2: Does the post include offensive or inappropriate language?

- **Yes:** Proceed to next step
- **No:** Select **NON** label and move to Step 4

Step 3: Is the offense in the post targeted?

- **Yes:** Select the most appropriate target:
    - **IND:** One or more individuals, not related to any specific group
    - **GRP1:** A group related to gender
    - **GRP2:** A group related to ethnic identity
    - **GRP3:** A group related to religious affiliation
    - **GRP4:** A group related to political affiliation
    - **OTH:** Other groups (organizations, events, etc.)
- **No:** Select **PROF** label and move to Step 4

Step 4: Was the labeling decision difficult?

- **Yes:** Select **X** label and move to next post
- **No:** Move to next post

Fig. 1. Annotation guidelines used in this study.

A total of 5202 posts were annotated in accordance with the labeling guidelines, and the distribution of these labels is presented in Table I. According to this distribution, 4119 posts contain at least one type of offensive language, while 934 posts do not include any offensive content. Additionally, there are 149 posts labeled as "X" because they were either not in Turkish or considered difficult to evaluate. These posts were excluded from the model development process.

Disagreements between annotators occasionally emerged during the annotation process. As a result, the dataset provides not only the final labels established by consensus but also the individual binary annotations (0: No, 1: Yes) from each annotator answering the question "Does the post contain any offensive language?" These binary annotations were provided independently by three annotators for each post. This approach allows researchers to analyze inter-annotator agreement and investigate subjective differences.

The annotators participated in the project voluntarily through personal connections with the authors. All annotators are native Turkish speakers with higher education backgrounds. To ensure the creation of a higher quality and more consistent dataset, a brief training session was provided prior to the annotation process. Sincere appreciation is extended to the annotators for their dedicated efforts in completing this time-consuming and meticulous process on a fully voluntary basis.

C. Data Analysis

This section analyzes the reliability and consistency of the binary annotations performed by three independent annotators. Additionally, it examines the agreement between the individual annotations of each annotator and the consensus labels determined collectively by the three annotators. Various metrics have been used to assess inter-annotator agreement.

Initially, the overall agreement among the three annotators was calculated using Fleiss' Kappa, yielding a value of 0.76. According to the criteria proposed by Landis and Koch [34], this score corresponds to a substantial level of agreement, indicating that the annotators were largely consistent in their assessments of whether the posts contained offensive language.

The pairwise agreements between annotators were evaluated using Cohen's Kappa coefficients, and the following values were obtained:

- Annotator 1 vs. Annotator 2: $\kappa = 0.693$
- Annotator 1 vs. Annotator 3: $\kappa = 0.787$
- Annotator 2 vs. Annotator 3: $\kappa = 0.804$

According to this classification, these values indicate a substantial level of agreement. Although there are some differences among annotators, overall there is strong consistency in their evaluations.

TABLE I. FINAL LABEL DISTRIBUTION BASED ON ANNOTATOR CONSENSUS

Label	Description	Count	Percentage (%)
X	Non-Turkish or unintelligible content	149	2.86
NON	Non-offensive content	934	17.95
PROF	Profanity or non-targeted offensive language	2695	51.79
IND	Offensive content targeting an individual	833	16.02
GRP1	Gender-based hate speech	38	0.73
GRP2	Ethnicity-based hate speech	12	0.23
GRP3	Religion-based hate speech	25	0.48
GRP4	Political affiliation-based hate speech	157	3.02
OTH	Other group-based hate speech (e.g., organizations, events)	359	6.90

In this analysis, the agreement between each individual annotator and the consensus label determined by the three annotators was also evaluated. The resulting Cohen's Kappa values are:

- Annotator 1 vs. Consensus: κ = 0.830
- Annotator 2 vs. Consensus: κ = 0.853
- Annotator 3 vs. Consensus: κ = 0.950

The high Cohen's Kappa values between annotators and consensus labels indicate that using multiple annotators minimizes subjective differences and results in a reliable dataset. This demonstrates the effectiveness of the multi-annotator method. These results strongly suggest that consensus-based labels provide a robust gold standard for training and evaluating models.

IV. EXPERIMENTAL SETUP

This section provides information about the experiments conducted for the automatic detection of offensive language in Turkish social media posts and evaluates the results of these experiments.

A. Task Objectives and Definitions

This part evaluates examples of offensive language in Turkish social media posts within the scope of two different classification tasks. The tasks examined are as follows:

- **Task 1: Offensive vs. Non-Offensive Classification**

This task focused on detecting whether a given X post includes offensive language. The following categories were used:

- Offensive (1): Posts include offensive language (e.g., insults, profanity, threats).
- Non-Offensive (0): Posts do not contain offensive language or harmful content.
- **Task 2: Targeted vs. Non-Targeted Classification**

This task was conducted exclusively on posts containing offensive language. The aim of this task is to determine whether offensive posts target a specific person, group, organization, or community. The categories used are:

- Targeted (1): Offensive posts target a specific person, group, or community.
- Non-Targeted (0): Offensive posts do not target any specific entity.

Table II presents information on the classification tasks carried out in this study; it outlines the defined classes for each task, the labels used to form these classes, and the corresponding number of samples per class.

TABLE II. LABEL DISTRIBUTION IN THE TOLID DATASET FOR EACH CLASSIFICATION TASK

Task	Class	Included Labels	Count
Offensive classification	Offensive (1)	PROF, IND, GRP1, GRP2, GRP3, GRP4, OTH	4119
	Non-offensive (0)	NON	934
Targeted classification	Targeted (1)	IND, GRP1, GRP2, GRP3, GRP4, OTH	1424
	Non-targeted (0)	PROF	2695

In this study, similar to the approach proposed by Çöltekin [25], two classification models are presented aiming to detect whether a post contains offensive language and whether the offensive language is directed at a specific target. This approach allows for a comprehensive analysis by capturing both the existence and direction of offensive language.

B. Selection of Transformer-Based Models

This study examines recent state-of-the-art NLP models to detect offensive language in Turkish social media posts. In line with this objective, the following three powerful transformer-based models trained on Turkish were selected:

- **BERTurk**: Schweter [35] introduced this model as a powerful Turkish language model based on the BERT architecture and trained exclusively on Turkish texts. The BERT architecture consists of 12 transformer layers, each with 768 hidden units, and approximately 110 million parameters. Considering the impact of case sensitivity on meaning in Turkish, the cased (case-sensitive) version of the model was selected for use.
- **ConvBERTurk**: Developed to reduce the computational cost of BERT, ConvBERTurk is a powerful language model based on the ConvBERT architecture [36] and trained exclusively on Turkish data. The ConvBERT model enhances efficiency and speed by integrating dynamic convolution layers that capture local dependencies, thereby reducing the computational burden of BERT's global self-attention mechanism. It shows strong performance on social media posts due to its efficiency and accuracy. In this study, the case-sensitive (cased) version was selected.
- **ELECTRA-Turkish:** This model is a language model built on the ELECTRA architecture [37], which was developed to reduce BERT's computational cost and accelerate the training process. It is trained exclusively on Turkish data. Instead of masked language modeling, it employs a discriminative pretraining approach that distinguishes real and fake generated tokens. In the experiments, the cased version of the model trained on a 35 GB Turkish dataset was used.

C. Experimental Settings

This section describes the technical setup and training processes of the transformer-based models used for the two classification tasks conducted on the TOLID dataset. In addition, detailed information is provided on the software and hardware environment, training configurations, durations, and technical challenges encountered during the experiments.

TABLE III. AVERAGE EPOCH DURATIONS AND TOTAL TRAINING TIMES OF THE MODELS

Task	Model	Average Time Per Epoch, Minutes	Total Training Time, Hours
Offensive classification	BERTurk	≈2.8	≈4.5
	ConvBERTurk	≈3.1	≈5
	ELECTRA-Turkish	≈3.3	≈5.5
Targeted classification	BERTurk	≈2.2	≈3.5
	ConvBERTurk	≈2.5	≈4
	ELECTRA-Turkish	≈2.7	≈4.5

The models were developed in Python using the Hugging Face Transformers library and trained on Kaggle Notebooks. The experiments were conducted on Kaggle's cloud-based notebook environment, which provides an Intel Xeon CPU, approximately 13 GB of RAM, 16 GB of GPU memory (NVIDIA Tesla P100), and a Linux-based operating system. The number of epochs was limited to 20, but early stopping enabled the models to converge within 8–10 epochs. The training configuration included a batch size of 16, a maximum input length of 128 tokens, and optimization with the AdamW algorithm combined with a linear learning rate scheduler. Class weights were incorporated into the loss function to mitigate class imbalance.

The TOLID dataset was evaluated using Stratified 10-Fold Cross-Validation. In each fold, 90% of the data was used for training and the remaining 10% for testing. Table III presents a summary of the training times recorded during the experiments. The Offensive vs. Non-offensive task required a longer training time due to its larger number of samples. Among the models, ELECTRA-Turkish has the longest training duration, followed by ConvBERTurk and BERTurk, respectively. The performance metrics reported in Tables IV and V represent the macro-averaged scores across all 10 folds.

Kaggle's free GPU support and pre-installed libraries made it possible to conduct experiments efficiently, although restrictions such as the nine-hour session limit and 16 GB of GPU memory imposed certain limitations. Nevertheless, this experimental setup provided the opportunity to train, test, and compare transformer-based models in detail for detecting offensive language in Turkish social media posts.

D. Experimental Results

A comprehensive evaluation and comparison of the performance of BERTurk, ConvBERTurk, and ELECTRA-Turkish models is provided in this section for the two classification tasks. The performance metrics—Macro F1-score, accuracy, precision, recall, and ROC-AUC (Receiver Operating Characteristic – Area Under the Curve) —were calculated through 10-fold cross-validation. Additionally, the results were compared with those reported in the literature, discussing the models' advantages and limitations.

The results of the experiments conducted for the Offensive vs Non-offensive classification task are shown in Table IV. In this task, the ConvBERTurk model achieved the highest performance, with a Macro F1 score of 0.827 indicating balanced success across classes. Additionally, the ROC-AUC values above 0.92 for all models highlight the strong capability in accurately identifying offensive posts.

Table V presents the performance results for the Targeted vs Non-Targeted classification task, which focuses on identifying whether offensive content is directed at a specific target. ConvBERTurk outperformed the other models in this task as well, though the margin was smaller. Overall, the performance scores were lower than those observed in the Offensive vs Non-offensive task. This may be due to the smaller dataset and the greater semantic ambiguity of the Targeted classes.

The study results indicate that all three transformer-based models effectively detect offensive language and identify its target in Turkish social media posts. ConvBERTurk outperformed BERTurk and ELECTRA-Turkish in both tasks, likely due to its CNN-based architecture that effectively captures local n-gram patterns and enhances contextual representations. Overall, the high ROC-AUC and Macro F1 scores demonstrate that all models perform well in detecting offensive language in Turkish and hold promise for deployment in real-world applications.

V. DISCUSSION

By introducing the TOLID dataset and applying transformer-based models, along with the achieved results, this study offers valuable contributions to Turkish offensive language detection research.

As discussed in the Introduction, prior Turkish offensive language datasets suffer from limited scope, class imbalance, and topic-specific focus [11–15, 25]. To address these gaps, the TOLID dataset was developed as an extensive resource that covers diverse forms of offensive language without being limited to specific topics, individuals, or groups. Its hierarchical annotation structure follows the framework proposed by Zampieri et al. [24] and is aligned with the guidelines from Çöltekin [25]. Furthermore, by applying classification algorithms during data collection, the dataset achieved a higher proportion of offensive content, ensuring a balanced distribution across both general categories and subcategories.

As summarized in Table VI, ConvBERTurk outperformed previous Turkish studies in the Offensive vs. Non-offensive classification task, achieving a Macro F1 score of 82.76% and an accuracy of 89.7%. This notable improvement is likely due to the balanced dataset, higher label quality achieved through multiple annotators, and the CNN-based architecture of ConvBERTurk, which better handles the short and informal nature of social media text.

In the Targeted vs. Non-targeted classification task, Table VI also shows that ConvBERTurk achieved the highest performance, reaching a Macro F1 score of 78.14% and an accuracy of 77.8%, slightly outperforming the F1 score of 77.9 reported by Çöltekin [25]. Nevertheless, the difference in performance among the models was relatively minor. This may be attributed to the greater semantic ambiguity of targeted offensive content and the requirement for deeper contextual information, both of which can limit the models' classification capabilities. Furthermore, the relatively low number of targeted instances in the dataset may also have constrained the models' performance.

Despite its contributions, this study has certain limitations. Firstly, relying solely on posts from the X platform could limit the models' ability to generalize to other social media platforms, given X's character limit,

TABLE IV. TEST PERFORMANCE OF MODELS ON THE OFFENSIVE VS NON-OFFENSIVE TASK

Model	F1 Score (Macro)	Accuracy	Precision	Recall	ROC-AUC
BERTurk	0.810	0.885	0.815	0.812	0.929
ConvBERTurk	0.827	0.897	0.833	0.824	0.938
ELECTRA-Turkish	0.808	0.884	0.813	0.807	0.926

TABLE V. TEST PERFORMANCE OF MODELS ON THE TARGETED VS NON-TARGETED TASK

Model	F1 Score (Macro)	Accuracy	Precision	Recall	ROC-AUC
BERTurk	0.777	0.790	0.774	0.791	0.858
ConvBERTurk	0.781	0.796	0.778	0.791	0.863
ELECTRA-Turkish	0.780	0.792	0.777	0.797	0.857

unique jargon, and informal language style. Secondly, as the dataset reflects posts from a specific time period, the models may underperform when encountering new slang, cultural trends, or political discourse that evolve over time. Thirdly, the relatively small number of samples in certain subcategories limits the models' effectiveness in fine-grained classification tasks. Finally, the substantial computational demands of transformer-based architectures pose challenges for real-time deployment on devices with limited resources.

Future studies could focus on overcoming current limitations by:

- Expanding the dataset with posts from platforms such as Instagram, TikTok, and Ekşi Sözlük to enhance diversity and generalizability.
- Increasing the number of samples in underrepresented subcategories to improve fine-grained classification performance.
- Incorporating continual learning techniques to help models adapt to the ongoing evolution of language and culture, including changes in slang and emerging trends.
- Developing lighter and faster models (e.g., DistilBERT, TinyBERT) suitable for real-time use on devices with limited resources.
- Developing multimodal approaches that combine text with accompanying images and videos shared in social media posts.
- Investigating how offensive language detection systems might affect freedom of speech and raise ethical concerns.

VI. CONCLUSION

This study presents a comprehensive framework for detecting offensive language in Turkish social media posts. The TOLID dataset, meticulously created by three independent annotators, includes not only general offensive expressions but also covers various subcategories such as sexist, racist, political, and religious insults, resulting in a balanced and reliable resource. By doing so, it addresses major limitations of previous research, including class imbalance and limited coverage, and makes a valuable contribution to Turkish NLP.

TABLE VI. PERFORMANCE COMPARISON OF PREVIOUS STUDIES AND THIS STUDY

Study	Task	Data	Model	F1
Çöltekin [25]	Offensive language detection	Turkish corpus	SVM	0.773
Çöltekin [25]	Target identification	Turkish corpus	SVM	0.779
Beyhan [15]	Sexism detection	Istanbul Conv.	BERTurk	0.779
Beyhan [15]	Racism detection	Refugee dataset	BERTurk	0.649
This study	Offensive language detection	TOLID	ConvBERTurk	0.827
This study	Target identification	TOLID	ConvBERTurk	0.781

SVM, support vector machine; TOLID, Turkish Offensive Language Identification Dataset.

The experiments with transformer-based models highlighted that the ConvBERTurk model achieved the highest performance. It outperformed prior work by reaching a Macro F1 score of 82.76% in detecting offensive language and 78.14% in distinguishing targeted from non-targeted offensive posts. These findings indicate that the combination of a detailed, multi-annotated dataset with modern deep learning architectures can effectively tackle the challenges inherent in Turkish social media texts.

Moreover, by introducing a web-based application, this study extends beyond academic research and provides a practical moderation tool suitable for real-world scenarios. In this way, detecting offensive language in Turkish social media posts acquires both scientific and social value.

In conclusion, through the introduction of the TOLID dataset, extensive experimentation, and the development of high-performing models, this study provides a solid foundation for future research on offensive language detection in Turkish. Further research may focus on expanding dataset size and diversity, tailoring models for various social media platforms, and developing multilingual systems that are sensitive to cultural context.

Data Availability Statement: The data that support the findings of this study are available on request from the corresponding author.

Artificial Intelligence Usage Statement: This study used ChatGPT (OpenAI, GPT-4) to improve English language and provide wording suggestions during the preparation of the manuscript.

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